

Remember that this formula still assumes that the investments are made once at the beginning of the multi-year tenure and returns are received once at the end of the multi-year tenure. The interest earned out of the investment, however, is re-invested and contributes to the overall returns.

Calculating CAGR

The CAGR is calculated using Formula 3:

Formula 3

$$\text{CAGR} = [(\text{Current market Value}) / (\text{Cost of Investment})]^{(1/n)} - 1$$

Where 'n' is the number of years for which the amount was invested. Let us use a modification of Example 2 to illustrate the concept of CAGR and how it is different from simple annualized returns.

Example 3

I invested in equity NFO two years back at ₹10 per unit (assuming no entry load). Today the NAV quoted for that mutual fund stands at ₹15 per unit. What is the annualized return on my invested amount if I were to assume that any interest earned in the first year is re-invested back in the second year?

Solution 3: Calculating Annualized Simple ROI

The CAGR calculation is based on Formula 3. The calculation would be as follows:

$$\begin{aligned}\text{CAGR} &= (15/10)^{(1/2)} - 1 \\ &= 1.5^{0.5} - 1\end{aligned}$$